

JAI BHAGAT

<https://jkbhagatio.io>
jkbhagatio[at]gmail[dot]com

SELECTED SKILLS

ML & AI

Mechanistic Interpretability
Deep & Reinforcement Learning
(incl. Multimodal Transformers,
LSTMs, CNNs, TD, DQNs, PPO)
Supervised Learning
(incl. GLMs, SVMs, Forests)
Unsupervised Learning
(incl. HDBSCAN, OPTICS)
Distributed Training
(incl. DDP, FSDP)

Programming Languages

Python (incl. PyTorch, Jax),
Bonsai, C, Rust, CUDA

Software Services

Wandb, Docker, Slurm,
AWS (EC2, ECS, S3), GCP, HF 🐙

Mechatronics

Micro-controllers -computers
(incl. Arduino, R Pi)
Simple PID, KF Control Systems
Electrophysiology Acquisition
CAD & 3d Printing
Laser Cutting

Wet Lab

In-vivo Electrophysiology
Genotyping
Optogenetics
Stereotaxic surgeries
Histology

SELECTED AWARDS

Bogue Fellowship 2024
Fondation JFMLCT 2023
UCL AWPO 2022
SWC Public Engagement Fund 2022
SWC Ph. D. Scholarship 2021

EXTRA TRAINING

MARS Scholar - AI Alignment
ARENA scholar - AI Alignment
Machine Learning Summer School
Cajal Ephys Acquisition Course
Advanced Study Program - MIT

EDUCATION

Ph. D. Computational Neuroscience | University College London 2025
B.A. Neuroscience | Boston University 2015

SELECTED PROFESSIONAL EXPERIENCE

Founding Machine Learning Researcher, [Metamorphic](#) 2026/03 – Present
Palo Alto, CA, USA
Machine Learning Research Scientist, [Enigma](#) 2025/09 – 2026/02
Palo Alto, CA, USA
Bogue Fellow Researcher, [UCL - Anthropic](#) 2025/01 – 2025/03
San Francisco, CA, USA
Software Developer, [CortexLab](#) & [International Brain Lab](#) 2018/10 – 2020/08
University College London, London, UK
Technical Associate I/II, [Wilson Lab](#) 2016/06 – 2018/06
Massachusetts Institute of Technology, Cambridge, MA, USA

SELECTED PUBLICATIONS

F. D'Agostino*, A. Gilbert*, S. Keller*, ... **J. Bhagat**, et al. Evaluating Neural Data Tokenizers: A Framework for Assessing Learned Representations of Spiking Activity. *In Review*.
J. Bhagat*, S. Molas-Medina*, G. Giglemiani, S. Heimersheim. [Compressed computation is not computation in superposition](#). *NeurIPS Mechanistic Interpretability Workshop* 2025.
J. Bhagat, A. G. Pouget, S. Molas-Medina. [A pipeline for interpretable neural latent discovery](#). *NeurIPS Data on the Brain & Mind Workshop* 2025.
D. Campagner* **J. Bhagat***, **G. Lopes***, et al. [Aeon: An open-source platform to study the neural basis of ethological behaviors over naturalistic timescales](#). *Biorxiv* 2025 & *In Press*.
International Brain Laboratory, K. Banga, J. Benson, **J. Bhagat**, et al. [Reproducibility of in-vivo electrophysiological measurements in mice](#). *eLife* 2025.
N. Steinmetz*, C. Aydin*, A. Lebedeva*, M. Okun*, M. Pachitariu*, **J. Bhagat**, et al. [Neuropixels 2.0: A high-density probe for stable, long-term brain recordings](#). *Science* 2021.
J. Bhagat*, M. J. Wells*, et al. [Rigbox: An open-source toolbox for probing neurons and behavior](#). *eNeuro* 2020.
J. Bhagat, et al. [LSTM neural networks for LFP event detection and classification in the rodent hippocampal-cortical network](#). *MIT BCS Symposium* 2018.

SELECTED OPEN-SOURCE PROJECTS

[Neuronauts](#): An educational outreach program for teaching teenagers fundamentals in engineering, computer science, neuroscience, and artificial intelligence. (*Founder*)
[nanoGPT](#): A minimal (nanomal?) Python repository containing code for building, training, and running nanoGPT. (*Sole creator, developer, maintainer*)
[Wall-E-GPT](#): Python and Arduino code for a GPT-controlled, semi-autonomous rover robot running on a Raspberry Pi. (*Sole creator, developer, maintainer*)
[aeon_mecha](#): [Project Aeon](#)'s main Python library for interfacing with acquired experiment data. (*Creator, developer, maintainer*)
[aeon_experiments](#): [Project Aeon](#)'s main Bonsai and C# library for running behavioral neuroscience experiment workflows. (*Developer, maintainer*)
[ibllib](#): The International Brain Laboratory's core shared Python libraries for data pipeline management and analysis. (*Developer, maintainer*)
[Rigbox](#): A MATLAB and C based toolbox for running behavioral neuroscience experiments and managing data. (*Developer, maintainer*)
[J.Clust](#). A complete, MATLAB spike sorting package. (*Sole creator, developer, maintainer*)